View Reviews

Paper ID
520

Paper Title
Multimodal Neonatal Pain Classification Using EEG, ECG, and Respiration Signals with Subject-Level Validation

Track Name
Special Track 2 [Advanced Computational Intelligence and Smart Digital Technologies (Optimization, AI & ML, Data Analytics, Cloud Intelligence, IoT, Cybersecurity, and Smart Applications)]

Reviewer #1

Questions

6. Comments to the Author(s) [Please provide a detailed assessment of the manuscript including the strengths and weaknesses of the paper in terms of novelty, technical content, relevance to Conference, quality of references, and experiments] Note: Your suggestions are extremely important for the authors to improve the quality of the paper. We humbly request you to provide feedback to authors.

The reported AUCs (0.849 ± 0.285; 0.929 ± 0.175) show substantial variability, yet no confidence intervals or statistical comparisons support the claims of competitive performance. F1-scores are also low and highly variable (0.475 ± 0.450; 0.452 ± 0.438), despite strong AUCs, requiring analysis of threshold selection or class imbalance. The small PIPP cohort (n=28) also limits LOSO generalisability.

Reviewer #3

Questions

6. Comments to the Author(s) [Please provide a detailed assessment of the manuscript including the strengths and weaknesses of the paper in terms of novelty, technical content, relevance to Conference, quality of references, and experiments] Note: Your suggestions are extremely important for the authors to improve the quality of the paper. We humbly request you to provide feedback to authors.

The manuscript presents a well-designed multimodal framework for neonatal pain classification with rigorous subject-level validation. The methodology is technically sound and the results are promising.

Minor improvements in external validation, model interpretability, and computational analysis would further strengthen the contribution.

Reviewer #4

Questions

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Improve the citations and add some recent one's too